

GPT-6.1 TERRA

10MAA

End of Year Revision Examination 2026

Paper 2

Student's full name: _____

Form class: _____

Maths teacher's code: _____

Time allowed: 2 hours

Instructions:

- Write in blue/black pen only. Use a pencil for diagrams.
- No twink or correction fluid allowed.
- Calculators are permitted.
- Answer **all questions in this booklet**, showing working in the appropriate space.
- Unless otherwise stated, give non-exact answers to **three significant figures**, angles to **one decimal place**, and money to **two decimal places**.

Topic	**Mark**
Number	/12
Algebraic Expressions & Equations	/32
Statistics & Probability	/26
Trigonometry	/16
Measurement	/15
Coordinate Geometry	/11
Graphs of Quadratic Functions	/20
Geometry	/20
Total	**/152**

Useful Formulae

The Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Volume of a cylinder

$$V = \pi r^2 h$$

Volume of a sphere

$$V = \frac{4}{3}\pi r^3$$

Volume of a cone

$$V = \frac{1}{3}\pi r^2 h$$

The Cosine Rule

$$a^2 = b^2 + c^2 - 2bc\cos A$$

The Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

The Area of a Triangle

$$A = \frac{1}{2}ab\sin C$$

The midpoint of a line segment

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Distance

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Gradient of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Equation of a line

$$y - y_1 = m(x - x_1)$$

Perpendicular criteria

$$m_1 m_2 = -1$$

For lines with defined gradients.

Number — 12 marks

1. A laptop's price is increased by 15% to $\$1,380$. Find its price before the increase.

Answer: ..... [3]

2. A savings account contains $\$2,400$. It earns 4% compound interest per year. Find the balance after 3 years.

Answer: ..... [3]

3. Write $0.\dot{3}\dot{6}$ as a fraction in its simplest form. Show your method.

Answer: ..... [3]

4. A program processes 3.2×10^6 records per hour. How many records does it process in 15 minutes? Give your answer in standard form.

Answer: ..... [3]

Algebra — 32 marks

1. Solve $5x - 7 = 2x + 11$.

Answer: ..... [2]

2. Solve $3(2t - 1) > 4t + 7$.

Answer: ..... [2]

Algebra continued

3. Simplify $\dfrac{(4a^3b^2)(3a^2)}{6ab}$, where $ab \neq 0$.

Answer: ..... [2]

4. Expand and simplify $(x-5)(x+3)$.

Answer: ..... [2]

5. Factorise fully $6x^2 - 24$.

Answer: ..... [2]

6. Simplify fully $\dfrac{x^2-9}{x^2+5x+6}$, stating the values of x excluded by the original expression.

Answer: ..... [3]

7. Make h the subject of $A = 2\pi r(r+h)$, where $r > 0$.

Answer: ..... [2]

8. A coding club buys 18 stickers. Linux stickers cost \$2 each and AI stickers cost \$3 each. The total cost is \$43. Find the number of each kind. Show your equations.

Answer: ..... [4]

9. Solve $x^2 - 7x + 10 = 0$.

Answer: ..... [3]

Algebra continued

10. A script processes x files per minute. An improved script processes $(x + 4)$ files per minute. Each processes 120 files. The improved script takes 1 minute less.

    (a) Write an equation in x describing this information.

    Answer: ..... [2]

    (b) Solve your equation to find x. Explain why any unsuitable solution is rejected.

    Answer: ..... [3]

11. A quadratic sequence starts $3, 8, 15, 24, 35, ...$ Find an expression for its nth term.

    Answer: ..... [3]

12. Write $x^2 - 6x + 2$ in completed square form.

    Answer: ..... [2]

Statistics and probability — 26 marks

1. A test recorded these build times in seconds: $4, 5, 5, 7, 9, 12, 14$.

   (a) Find the median and the mean.

   Answer: ..... [3]

   (b) The slowest result was actually 21 seconds. State the new median and explain your answer.

   Answer: ..... [2]

2. The five-number summary of the build times, in seconds, for 20 Linux computers is: minimum 4, lower quartile 6, median 8, upper quartile 11, maximum 16.

   (a) Draw a box plot for these data on a labelled scale.

   Answer: ..... [3]

   (b) Find the interquartile range and explain what it measures.

   Answer: ..... [2]

3. Of 50 programmers, 30 use Linux, 27 use Python, and 5 use neither.

   (a) Find how many use both. You may draw a Venn diagram.

   Answer: ..... [3]

   (b) A programmer is chosen at random from those who use Linux. Find the probability that they also use Python.

   Answer: ..... [2]

4. A security test contains 5 safe and 3 malicious sample files. Two are selected at random without replacement.

(a) Find the probability that both are malicious.

Answer: ..... [2]

(b) Find the probability that exactly one is malicious.

Answer: ..... [3]

(c) Given that exactly one is malicious, find the probability that the first file selected was malicious.

Answer: ..... [2]

5. A grouped table records response times t in seconds.

$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 40$	$40 < t \leq 60$
4	8	12	6

(a) Estimate the mean response time.

Answer: ..... [3]

(b) Find the frequency density for $20 < t \leq 40$.

Answer: ..... [1]

Trigonometry — 16 marks

1. A cable runs from the top of a 12 m mast to a point 9 m from its base on level ground.

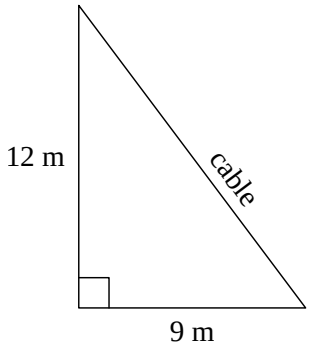


Diagram is NOT to scale

(a) Find the cable's length.

Answer: ..... [2]

(b) Find the angle the cable makes with the ground.

Answer: ..... [2]

Trigonometry continued

2. A robot travels 8 km due east from A to B, then 6 km on a bearing of 030° from B to C.

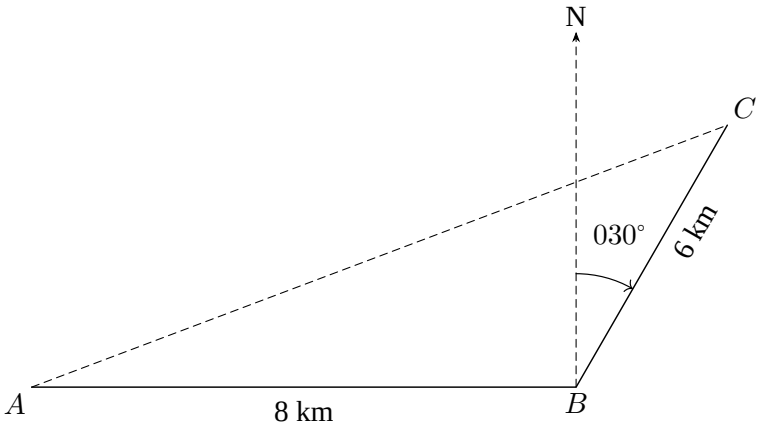


Diagram is NOT to scale

(a) Find angle ABC.

Answer: ..... [2]

(b) Calculate the distance AC.

Answer: ..... [3]

(c) Calculate the area of triangle ABC.

Answer: ..... [3]

Trigonometry continued

3. (a) Sketch $y = \sin\theta$ for $0° \leq \theta \leq 360°$. Label the axes and the coordinates of the intercepts and turning points.

Answer: ..... [3]

(b) Use the graph to solve $\sin\theta = 0.5$ for $0° \leq \theta \leq 360°$.

Answer: ..... [1]

Measurement 15 marks

1. Two similar solid cones have heights 6 cm and 15 cm. The smaller cone has volume 40cm^3. Find the volume of the larger cone.

Answer: ..... [3]

2. A cylindrical server cooling tank has radius 0.4 m and height 1.2 m.

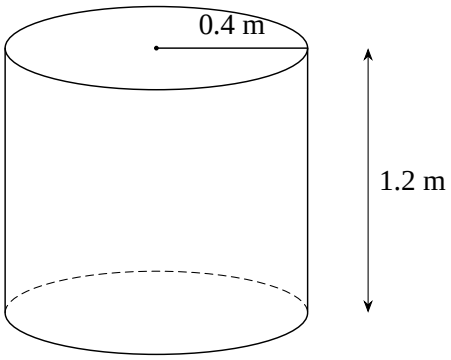


Diagram is NOT to scale

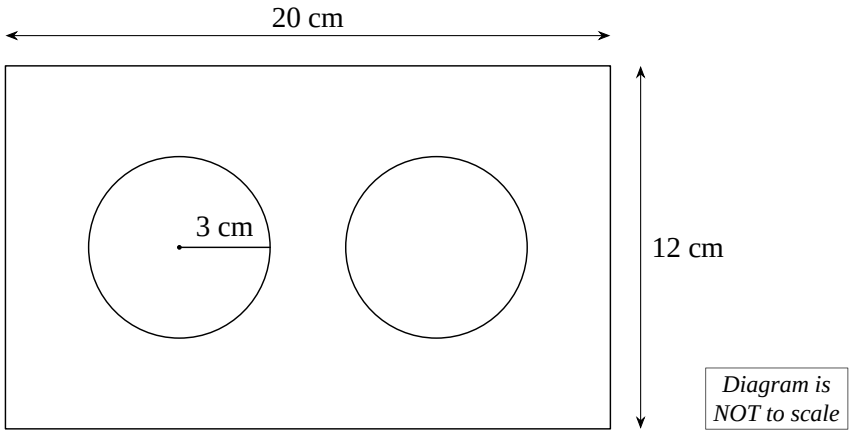
(a) Calculate its volume in cubic metres.

Answer: ..... [2]

(b) One litre is 1000cm^3. Find the capacity of the tank in litres.

Answer: ..... [2]

3. A rectangular metal plate measures 20 cm by 12 cm. Two circular holes, each of radius 3 cm, are cut from it. Find the remaining area, to the nearest square centimetre.



Answer: ..... [3]

4. A rectangular device case is 15 cm long and 8 cm wide, each correct to the nearest centimetre.

(a) State the lower and upper bounds for its length.

Answer: ..... [2]

(b) Find the greatest possible area to which the case's top can be arbitrarily close.

Answer: ..... [3]

Coordinate geometry — 11 marks

1. The points $A = (-2, 3)$ and $B = (4, 7)$.

   (a) Find the gradient of AB.

   Answer: ..... [2]

   (b) Find the midpoint of AB.

   Answer: ..... [1]

   (c) Find the length of AB, giving an exact answer.

   Answer: ..... [2]

   (d) Find the equation of the perpendicular bisector of AB in the form $y = mx + c$.

   Answer: ..... [4]

2. Determine whether $(10, -3)$ lies on $y = 2x - 23$. Give a reason.

   Answer: ..... [2]

Quadratic functions — 20 marks

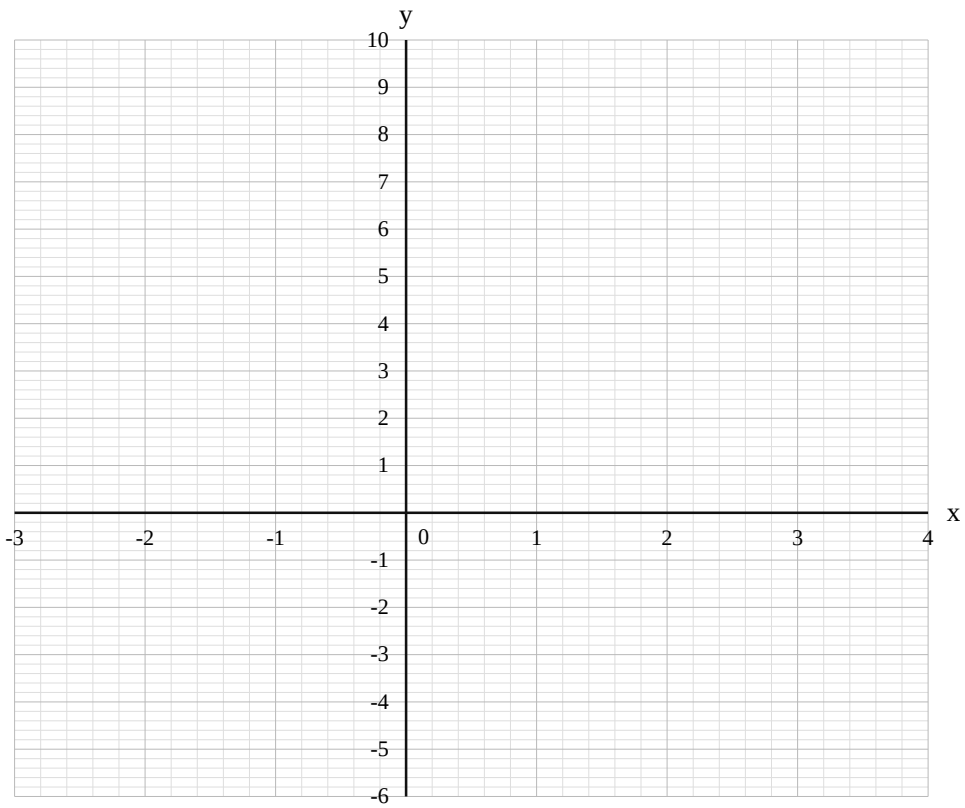
1. Let $f(x) = x^2 - 2x - 3$.

(a) Complete the table.

x	-2	-1	0	1	2	3
$f(x)$						

[2]

(b) On the axes below, plot these points and draw a smooth curve for $-2 \leq x \leq 3$.



[3]

(c) Find the coordinates of the turning point of f.

Answer: ..... [2]

(d) Solve $f(x) = 0$ algebraically.

Answer: ..... [2]

(e) Describe the transformation mapping $y = x^2$ to $y = (x - 1)^2 - 4$.

Answer: ..... [2]

2. The height h metres of a drone above level ground after t seconds is modelled by $h = (t + 1)(7 - t)$ for $0 \leq t \leq 7$.

(a) Find its height at $t = 0$.

Answer: ..... [1]

(b) When does the drone reach the ground?

Answer: ..... [1]

(c) Find its maximum height and the time at which this occurs.

Answer: ..... [3]

(d) Find both times when the drone is 12 m above the ground.

Answer: ..... [4]

Geometry — 20 marks

1. In triangle ABC, $AB = AC$ and angle $BAC = 38°$. Find angle ABC, giving a geometric reason.

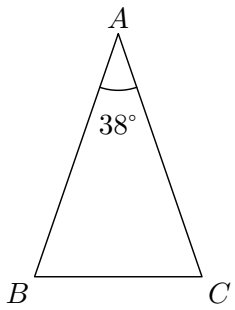


Diagram is NOT to scale

Answer: ..... [2]

2. Quadrilateral $PQRS$ is cyclic. Angle $PQR = 112°$.

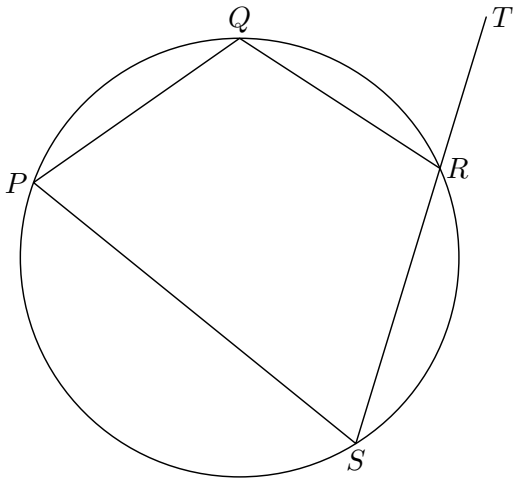


Diagram is NOT to scale

(a) Find angle PSR, giving a reason.

Answer: ..... [2]

(b) Side SR is extended beyond R to T. Find angle QRT if angle $QPS = 74°$. Give reasons.

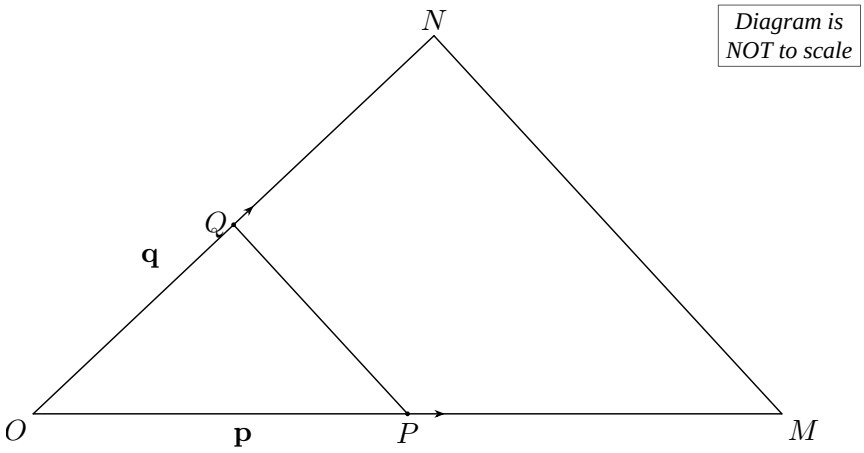
Answer: ..... [2]

Geometry continued

3. Draw two points A and B 6 cm apart. Using a ruler and compasses, construct the locus of points equidistant from A and B. Leave your construction arcs visible.

Answer: ..... [3]

4. In triangle OMN, $\overrightarrow{OM} = \boldsymbol{p}$ and $\overrightarrow{ON} = \boldsymbol{q}$. P is the midpoint of OM, and Q is the midpoint of ON.



(a) Express $\overrightarrow{MN}$ in terms of $\boldsymbol{p}$ and $\boldsymbol{q}$.

Answer: ..... [1]

(b) Express $\overrightarrow{PQ}$ in terms of $\boldsymbol{p}$ and $\boldsymbol{q}$.

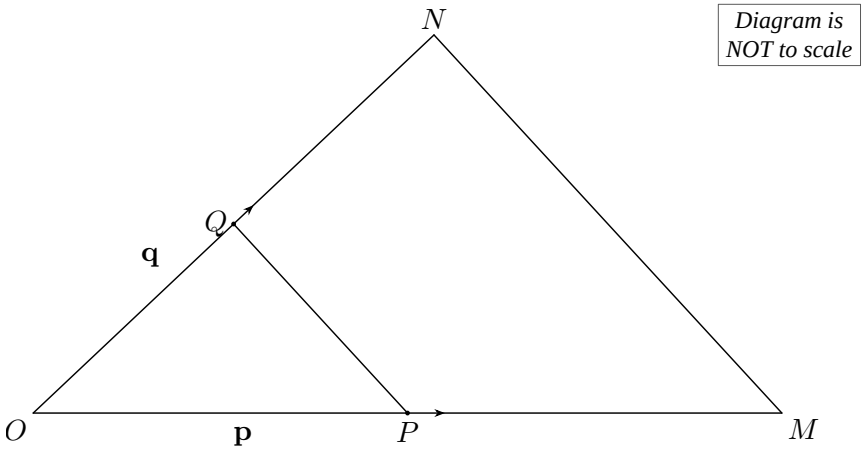
Answer: ..... [2]

(c) Prove that PQ is parallel to MN and half its length.

Answer: ..... [3]

5. In triangle OMN, $\overrightarrow{OM} = \boldsymbol{p}$ and $\overrightarrow{ON} = \boldsymbol{q}$. P and Q are the midpoints of OM and ON respectively. A point R has position vector $\overrightarrow{OR} = \frac{1}{3}\boldsymbol{p} + \frac{1}{3}\boldsymbol{q}$.

The position of R is not shown on the diagram.



(a) Find $\overrightarrow{PR}$ in terms of $\boldsymbol{p}$ and $\boldsymbol{q}$.

Answer: ..... [2]

(b) Show that R lies on PN, and find the ratio $PR : RN$.

Answer: ..... [3]

End of assessment